EJIN HORO BANNER MSW LANDFILL BIOGAS POLLUTION CONTROL AND COMPREHENSIVE UTILIZATION PROJECT

Project title	Ejin Horo Banner MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project
Project ID	5344
Crediting period	09-Dec-2023 to 31-Dec-2028
Original date of issue	13-Nov.-2024
Most recent date of issue	13-Nov.-2024
Version	1.0
VCS Standard Version	4.7
Prepared by	Jie Zhang, Henan BCCY Environmental Energy Co., Ltd.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Ejin Horo Banner MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project (hereafter referred to as the project) is located at MSW landfill site, #4 Community, Zhang Gangtu Village, Altenheige Town, Ejin Horo Banner, Ordos City, Iner Mongolia Autonomous Region, PR China. The project is invested, constructed and operated by Ordos City BCCY New Energy Co., Ltd. The objective is to collect the landfill gas (LFG) from Ejin Horo Banner MSW Landfill site (hereafter referred to as Ejin Horo Banner landfill site) and utilize it to generate electricity.

Ejin Horo Banner landfill site is harmless treatment landfill site with a total design effective storage capacity of1,800,000 m³, design daily municipal solid waste (MSW) disposal of 600 tons. At present, the daily disposal is around 380 tons. Ejin Horo Banner landfill site is expected to close at the end of 2028[1]. The project has a total designed capacity 2MW (2*1000kW) and combusts the LFG, which contains nearly 50% of methane, to generate electricity and export it to the North China Grid Company (NCGC). The project started construction on 11-Aug-2023 and started operation on 09-Dec-2023. The crediting period of the project started on 09-Dec-2023 and will end on 31-Dec-2028[2].

The project collects and utilizes LFG from Ejin Horo Banner MSW landfill site for power generation. Otherwise, the LFG would be released into the atmosphere. Methane in the LFG has been emitted to the atmosphere since the landfill started operation. Therefore, the project avoids methane emissions, and, at the same time, electricity is generated from a renewable energy source, instead of being provided by the grid.

Scenario existing prior to the start of the implementation of the project activity (the same as baseline scenario)

- LFG from Ejin Horo Banner MSW landfill site is released into the atmosphere.
- Equivalent electricity generated by the project is supplied by the NCGC, which is dominated by fossil fuel based power plants.

[1] Source: FSR of Ejin Horo Banner Landfill site

[2] Source: FSR. It is the expected closure time of the landfill site. The closure time might very due to the amount of garbage entering the landfill site.

The total power generation during the operation period is estimated to be 34,775 MWh. The estimated average GHG emission reduction of the project is 42,936 tCO2e and the total GHG emission reductions and removals are 217,269 tCO2e during the crediting period 10-Dec-2023 to 31-Dec-2028 (first and last days included).

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	09-December-2023 to 31-December-2028	VCS	LGAI Technological Center, S.A. (Applus+ Certification)	5years23 days

1.3 Sectoral Scope and Project Type

The project is a non-AFOLU project and not a grouped project activity.

Sectoral scope[3]	1: Energy industries (renewable-/non-renewable sources) 13: Wastehandling and disposal
Project activity type	The project is neither a AFOLU project nor a grouped project

The project is an LFG recovery to power project and not an AFOLU project. Therefore, the following table isn't applicable to the project.

Sectoral scope	N/A
AFOLU project category[4]	N/A
Project activity type	N/A

1.4 Project Eligibility

1.4.1 General eligibility

As per section 2.1.1 of VCS Standard (version 4.7), the scope of the VCS Program includes:

VCS Standard Scope	Fulfilled (Yes/No)	Justification

[3] Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes

[4] See Appendix 1 of the VCS Standard

(1) The seven Kyoto Protocol greenhouse gases.	Yes	The project is an LFG recovery to power project. Thus, the project is applicable to this scope.
(2) Ozone-depleting substances (ODS).	N/A	The project activity does not involve any Ozone depleting substances.
(3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	N/A	The project activity is supported by CDM methodology.
(4) Project activities supported by a methodology approved under an approved GHG program, unless	Yes	The applied methodologies ACM0001 (Version 19.0) of the project are methodologies approved
(5) Jurisdictional REDD+	N/A	This is not a Jurisdictional REDD+

Furthermore, the project does not belong to the projects excluded in Table 1 of VCS Standard (Version 4.7).

Thus, the project is eligible under the scope of VCS program.

According to 3.1 of VCS standard (Version 4.7), establishing a consistent and standardized certification process is critical to ensuring the integrity of VCS projects. Accordingly, certain high-level requirements must be met by all projects, as set out below.

Reference to VCS	Requirements	Justification
3.1.1	Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1	The project meets all applicable rules and requirements set out under the VCS Program, including VCS standard (version 4.7).

3.1.2	Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1. The list of methodologies and their validity periods is available on the Verra website.	The project applies the methodology ACM 0001- Flaring or use of landfill gas (Version 19.0) along with tools or modules as applicable developed under the United Nations Clean Development Mechanism, which can be used for projects and programs registering with VCS as per requirements available on the Verra website.
3.1.3	Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.22 below. Projects shall update to the latest version of the methodology when reassessing the baseline or renewing a crediting period.	The project applies to the methodology ACM0001-Flaring or use of landfill gas (Version 19.0), which are the latest version.
3.1.4	Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether the law is enforced.	According to the environment impact assessment report (EIA) of the project, the project complies with all Chinese relevant laws and regulations.
3.1.5	3.1.5 Where projects apply methodologies that permit the project proponent its own choice of model (see the VCS Program Definitions for the definition of model), the model shall meet the requirements set out in the VCS Methodology Requirements, and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or carbon dioxide removals).	Not applicable as there is no model that needs to be chosen by the project proponent as per the applied methodology ACM0001-Flaring or use of landfill gas (Version 19.0).

3.1.6	Where projects apply methodologies that permit the project proponent to choose a third- party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet the requirements set out in the VCS Methodology Requirements.	Not applicable as there are no third-party default factor or standard need to be chosen by the project proponent as per the applied methodology ACM0001-Flaring or use of landfill gas (Version 19.0).
3.1.7	Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.	Not applicable. The project obeys the rules and requirements of the VCS Program
3.1.8	Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits (see the VCS Program Definitions for the definition of capacity limit) and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	Not applicable as there is no specified capacity limits or any other relevant requirements set out as per the applied methodology and tools.
3.1.9	Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 Document History and Effective Dates or equivalent for other program documents and are listed in a companion Summary of Effective Dates document which corresponds with each update.	The project has followed the latest requirements of Verra.

1.4.2 AFOLU project eligibility

The project is not an AFOLU project. Thus, this section is not applicable.

1.4.3 Transfer project eligibility

The project is not a transfer project. Thus, this section is not applicable.

1.5 Project Design

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

The project is not a grouped project. Thus, this section is not applicable.

1.6 Project Proponent

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Minister of carbon emission department
Address	10th Floor, Kineer Mansion, Ruyi River 2nd West St., Ruyi West Rd., Zhengdong New District, Zhengzhou City
Telephone	+86 371 65521780
Email	lwang@bccynewpower.com

1.7 Other Entities Involved in the Project

Organization name	Ordos City BCCY New Energy Co., Ltd.
Role in the project	Project Operation Entity
Contact person	Jie Zhang
Title	Project Manager
Address	10th Floor, Kineer Mansion, Ruyi River 2nd West St., Ruyi West Rd., Zhengdong New District, Zhengzhou City
Telephone	+86 371 56737901
Email	greatjenny@bccynewpower.com

1.8 Ownership

The project owner is Ordos City BCCY New Energy Co., Ltd. The approval of the Feasibility Study Report (FSR) and Environmental impact Assessment (EIA), and the business license of the project owner are the evidence for legislative rights. Besides, the equipment purchasing contract, and the purchasing power agreement are the evidence for the ownership of the plant equipment and power generating. Ordos City BCCY New Energy Co., Ltd. is a wholly owned subsidiary of Henan BCCY Environmental Energy Co., Ltd., which is responsible for the acquisition of approvals, construction and operation of project. Henan BCCY Environmental Energy Co., Ltd. is authorized as the project Proponent by Ordos City BCCY New Energy Co., Ltd.

1.9 Project Start Date

Project start date	09-Dec-2023
Justification	The project started to operate on 09-Dec-2023, which is lined with the requirement of the VCS standard (version 4.7).The project start date of a NON-AFOLU project is the date on which activities that lead to the generation of reductions or removals are implemented.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	10-Dec-2023 to 31-Dec-2028

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)

12-Dec-2023 to 31-Dec-2023	2,439
01-Jan-2024 to 31-Dec-2024	39,749
01-Jan-2025 to 31-Dec-2024	44,496
01-Jan-2026 to 13-Dec-2026	45,547
01-Jan-2027 to 13-Dec-2027	45,173
01-Jan-2028 to 13-Dec-2028	42,304
Total estimated ERRs during the first or fixed crediting period	219,708
Total number of years	5 years 23 days
Average annual ERRs	43,394

1.12 Description of the Project Activity

GHG emission reductions will be achieved through combustion of the recovered methane gas by gas engines, which would be otherwise emitted to the atmosphere, and the generation of electricity from the landfill gas, which is supplied by the China Southern Power Grid (CSPG) prior to the implementation of the project.

Description of the technology in the project

The project consists of LFG collection, pre-treatment system, with subsequent electricity generation and grid connection system.

LFG collection system

LFG is extracted from MSW under negative pressure by blower pumps and moved through wells, then collected by gas collection stations and transferred by sub-pipes and a main pipe to LFG treatment equipment. Flow rate of the LFG is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

LFG pre-treatment system

Prior to electricity generation, LFG is pre-treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturizing, cooling and pressurization.

Electricity generation and grid connection system

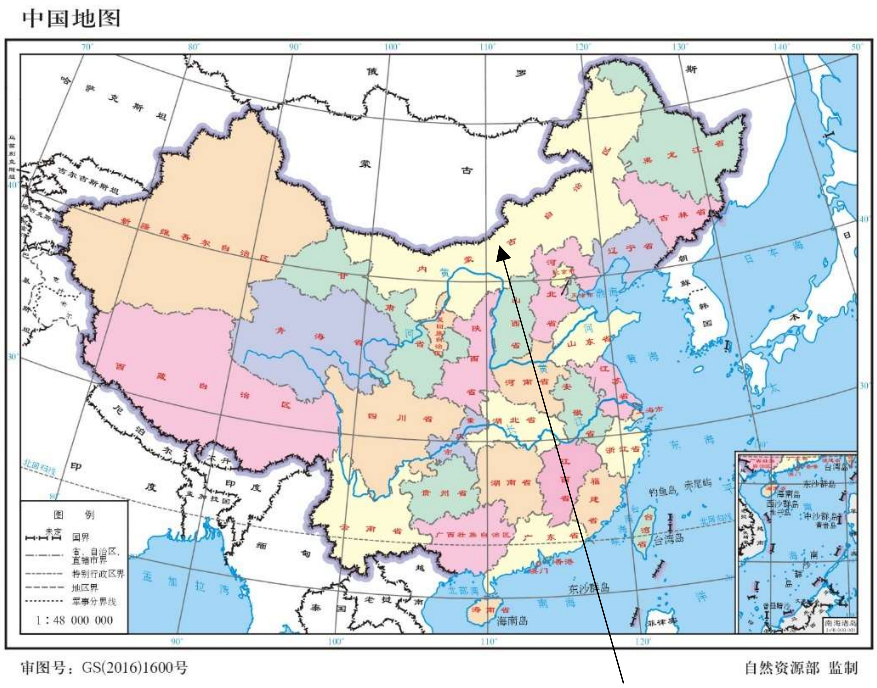
2 gas engines of 1MW rated power each are fed with LFG and generate electricity, which is then exported to the grid.

Table 1.1 Equipment technical parameters

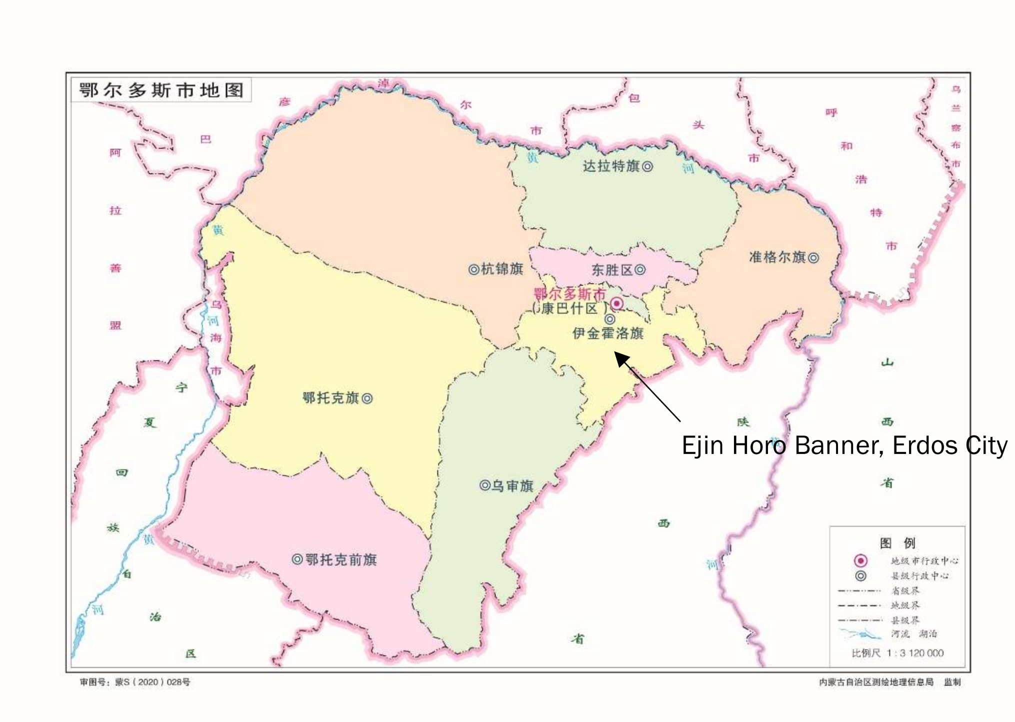
Main Equipment	Parameter	Value
Generator set	Type	1000GF-NK
	Manufacturer	CNPC Jichai Power Company Limited
	Rated capacity	1000 kW
	Rated rotation speed	1000 r/min
	Capacity factor	0.8 (lagging)

1.13 Project Location

The project is located at MSW landfill site, #4 Community, Zhang Gangtu Village, Altenheige Town, Ejin Horo Banner, Ordos City, Iner Mongolia Autonomous Region, PR China. The central geographical coordinates of the project site are longitude 109°36′55.95″E and latitude 39°32′42.54″N. Figure 1.1 shows the location of the Project.



Iner Monggol Autonomous Region



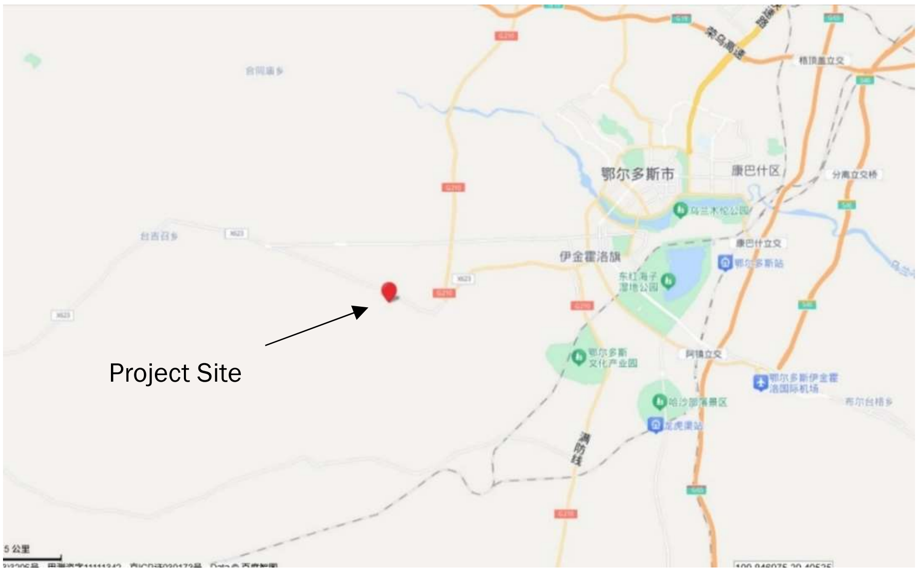


Figure 1.1 Location of the project

1.14 Conditions Prior to Project Initiation

The baseline scenarios are the same as the conditions existing prior to the project initiation. The details are shown in Section 3.4.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project is in the field of waste handling and disposal and meets the relevant provision of the law of the People's Republic of China on the Prevention and Control of Environmental Pollution by Solid Waste. The generation of power exported to NCGC will meet the requirement of national laws and regulations, also financially viable.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the VCS Program Definitions for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planned to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.18 Sustainable Development Contributions

Ejin Horo Banner MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project (hereafter referred to as the project) is located at MSW landfill site, #4 Community, Zhang Gangtu Village, Altenheige Town, Ejin Horo Banner, Ordos City, Iner Mongolia Autonomous Region, PR China. The project objective is to collect the landfill gas (LFG) from Ejin Horo Banner MSW Landfill site and utilize it to generate electricity. The project involves the installation of LFG collection, pre- treatment system, with subsequent electricity generation and grid connection system.

The project will contribute to Sustainable Development Goals in the following aspects:

SDG 8 Decent works and Economic Growth

The project can provide full time job opportunities over the project lifetime.

SDG 7 Affordable and Clean Energy

The project utilizes the LFG from the Ejin Horo Banner landfill site to generate electricity by gas engines. Total electricity generation supplied by the project to the grid is estimated to be 34,794MWh during the crediting period, which could replace the equivalent electricity from coal-based grid at the same time.

SDG 13 Climate Action

The project utilizes the LFG for electricity generation, which contains nearly 50% of methane, to substitute the equivalent electricity supplied by the grid. Total GHG emission reductions and removals of 219,708 tCO_2e will be achieved during the fixed crediting period.

Sustainable development priorities are illustrated in the People's Republic of China National Report on Sustainable Development issued on 01-Jun-2012. Sustainable development priorities of the Report related to the project are promoting new energy development, strengthening environmental pollution control and responding to climate change. Methane is an ideal clean fuel. Each cubic meter of methane contains about 360000kJ calorific value, LFG recovery and utilization will supplement the energy supply of the grid and contributes to the new energy development in China. Odor from landfill negatively impacts on residents around the landfill. Implementation of the project will reduce odors through LFG collection and thus mitigate the impact of landfill odor on people's daily lives. Landfill gas contains nearly 50% of methane and methane is a greenhouse gas some 28 times more potent than carbon dioxide. LFG recovery and utilization is contribution to mitigating global warming.

Table1.2: Sustainable Development Contributions

SDG Target	SDG Indicator	Project activity contribution
13.0	Tons of greenhouse gas emissions avoided or removed	The project avoided anthropogenic emissions of greenhouse gases (GHG) of 219,708 tCO_2e during the crediting period (09-Dec-2023 to 31-Dec-2028).
8.5	Total number of job opportunities	The project activity employs 10 permanent and full-time employees through its operational company Ordos City BCCY New Energy Co., Ltd. during the crediting period (09-Dec-2023 to 31- Dec-2028).
7.2	MWh of renewable energy generated	Total 34,794 MWh of renewable energy generated during the crediting period (09-Dec-2023 to 31-Dec-2028).

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not applicable as not AFOLU project.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further information is available.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The purpose of the project is to utilize LFG to produce electricity, which is supplied to NCGC. The entire project must adhere to local regulations, rules, and obtain necessary permissions. The following steps were adapted to identify stakeholders (based on Section 3.18 of VCS Standard—version 4.7). Step 1: List all individuals or groups who may have an impact on or be affected by the project. Step 2: Conduct a well-being ranking of local or community stakeholders. Step 3: Analyze the interests, motivations for participation, and relationships with other stakeholders of each stakeholder group. Step 4: Analyze the influence and importance of each potential stakeholder group. Step 5: Publish the results of the identified stakeholders. If anyone else believes they are one of the project's stakeholders, they can directly contact the project supporters for consultation. The final stakeholders of this project are as follows: 1. residents 2. local authorities 3. employees of Ejin Horo solid waste management station 4. employees of Ordos City BCCY New Energy Co., Ltd. 5. the project proponent
Legal or customary tenure/access rights	The project is located at Ejin Horo Banner landfill site and does not involve land issues.

	According to the project cooperation agreement signed between the project proponent and Inner Mongolia Tehong Environmental Protection Technology Co., Ltd. (the operator of Ejin Horo Banner landfill site), the project proponent has the right to use their LFG for power generation.
Stakeholder diversity and changes over time	Stakeholders identified for this project activity are diverse in nature in terms of socio-economic status, age and gender wisdom and the diversity is unlikely to change over time.
Expected changes in well-being	1. Economic well-being：The project activity is likely to provide employment for the locality thus economic well-being is expected. 2. Environmental well-being： (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it willbring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid isestimated to be 34,794 MWh, which could replace the equivalent electricity from coal-based grid.
Location of stakeholders	The location of stakeholders is #4 Community, Zhang Gangtu Village, Altenheige Town, Ejin Horo Banner, Ordos City, Iner Mongolia Autonomous Region, PR China
Location of resources	The location of resources is at Ejin Horo Banner landfill site, #4 Community, Zhang Gangtu Village, Altenheige Town, Ejin Horo Banner, Ordos City, Iner Mongolia Autonomous Region, PR China

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	15-Sep-2023
Stakeholder engagement process	The stakeholders were informed about the stakeholder meeting through posters on 14-Aug-2023. In the bulletin, the company invited the potential stakeholders to get to know their opinions and/or suggestions about the implementation of the project and VCS application of theproject. The stakeholder consultancy meeting for parties interested in the project was organized at Ejin Horo Banner landfill site on 15-Sep-2023 to collect opinions from all

	stakeholders, such as representatives of residents and local authorities, Ejin Horo Banner solid waste management station, the project proponent and so on. The project proponent appointed one person to make a meeting minute. Bulletin and meeting are in Chinese. The stakeholder meeting was attended by 15 people, six of whom were women.
Consultation outcome	During the stakeholder consultancy meeting, some stakeholders worried that the project would bring noise, land occupying and pollution on employees and residents' living. For these issues, the project proponent will offer some measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents.
Ongoing communication	Expecting to introduce the project information, grievance mechanism was also explained during the local stakeholder consultation meeting. A grievance expressionbook will be put in the office of the project, which is used tocollect stakeholders' views about the project. Project proponent will be checking the comments in the book on aregular basis and recording responses. The project information with contact information has also been posted on the bulletins at and near the project site. Every stakeholder can make comments by grievance expressionbook or contact information. Project Participant will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible.
Stakeholder input	The project did not receive major negative input during the consultation. The project design does not need to have any updates. All stakeholders were pleased with the development of the project. The project would facilitate the development of the local economy and increase the income of local residents.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Based on the principle of friendly consultation and on the basis of truly and fully expressing their respective wishes, the project proponent and Inner Mongolia Tehong Environmental Protection Technology Co., Ltd (the operator and manager of Ejin Horo Banner landfill site) signed a cooperation agreement on this project The feasibility study report of the project was approved by the Ejin Horo Banner Energy Bureau on 08-Nov-2023. An environmental impact assessment (EIA) report of the project was approved by Ordos City Ecological Environment Bureau Ejin Horo Banner Branch on 12-Oct-2023. According to stakeholder consultation meetings, residents have expressed support for the project. In general, the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), localcommunities (LCs), and customary rights holders. There are not ongoing or unresolved conflicts during theproject validation period and monitoring period.
Outcome of FPIC	The feasibility study report of the project was approved by the Ejin Horo Banner Energy Bureau on 08-Nov-2023. An environmental impact assessment (EIA) report of the project was approved by Ordos City Ecological Environment Bureau Ejin Horo Banner Branch on 12-Oct-2023. The project is located at Ejin Horo Banner landfill site, Gully One,Taxing Village, Altenheige Town, Ejin Horo Banner, Ordos City, Iner Mongolia Autonomous Region, PR China. According to the ecological environment control zoning map of Ordos City[5], the project is not located in nature reserves, scenic spots, drinking water source protection areas and other areas requiring special protection. After the operation of the project, the prevention measures proposed in the report will not affect the ecological environment and meet the requirements of the red line of ecological protection. Based on questionnaire surveys and stakeholder consultation meetings, residents have expressed support for the project.

[5] Source: Ordos Ecological Environment Bureau: https://sthjj.ordos.gov.cn/xwzx_6562/gsgg/202408/t20240809_3644598.html

	The project was constructed with the consent of government departments at all levels. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Development process	If there are conflicts of interest and dissatisfaction among stakeholders, they can choose to directly appeal to the project proponent or initiate an appeal meeting, which is usually the most effective way to solve the problem. Stakeholders can file appeals or complaints during the meeting through the phone number of the relevant contact person assigned by the project proponent. In addition, the progress of the project is made public on the VCS Pipeline website, accepting public feedback. If stakeholders have appeals and complaints, they can directly provide feedback to Verra. Their opinions will be taken into consideration, and the project proponent will promptly respond and provide feedback to Verra. In terms of government regulation, sometimes complaint mechanisms at the government level can also serve as an effective solution. This project is supervised by the local Ecological Environment Bureau, Development and Reform Commission, and Human Resources and Social Security Bureau. The above institutions all have public complaint telephone channels, through which stakeholders can provide feedback on their opinions. The project proponent will respond to it and provide feedback on improvement to government regulatory agencies. In the project, the project proponent has designated a staff member to record and collect dissatisfaction and complaints related to conflicts of interest, who play an important role in handling these common conflicts and appeals. Once a report regarding the appeal is received, theproject proponent will contact and discuss with relevantcommunities or other stakeholders within 24 hours striving to propose a solution and mediation plan within a week. In any of the above channels, once the project proponent receives a report on the appeal, the project proponent should contact and discuss with the relevant community orother stakeholders within 24 hours, striving to propose asolution and mediation plan within

	one week. If the problemcannot be resolved within 30 days, it is recommended that stakeholders consider filing a lawsuit.
Grievance redress procedure	The above appeal and compensation procedures were concluded through joint discussions with stakeholders at a stakeholder meeting.

During the stakeholder meeting, we have told the stakeholders about the grievance procedure of the project. They all agreed with the grievance procedure and felt the procedure was easily accessible to them for ongoing consultation.

2.1.5 Public Comments

This project is in the public comment period and no comments have been received.

Comments received	Actions taken
/	/

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project proponent has successfully operated several recoveries to power plants in China. It has also successfully developed carbon projects related to LFG recovery to power under VCS program (VCS reference number: 2353, 2731, 2840). Therefore, the PP has expertise and experience in implementing similar project activities.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	1. Economic well-being : the project activity is likely to provide employments to the locality thus economic well beings are expected. 2. Environmental well-being : (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity supplied by the project to the grid is estimated to be 115,895 MWh,

		which could replace the equivalent electricity from coal-based grid. The investment, construction and operation of the project is in compliance with the relevant laws and regulations of the People's Republic of China. Thus, there are no natural and human-induced risks to stakeholders' wellbeing.
Risks to stakeholder participation	No risk identified	Based on the stakeholder meeting, the stakeholders involved in the project are very supportive of the project. The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China.
Working conditions	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. The Project has DCS system. Therefore, most of the employees work in indoor environment (at the office), instead of having to stand outside. In case of on- site monitoring and device maintenance, the project proponent provides safety protection equipment.
Safety of women and girls	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. China has published "Women's Rights Protection Law" [6] to provide support and protection to women and girls.

[6] https://www.gov.cn/guoqing/2021-10/29/content_5647634.htm

Safety of minority and marginalized groups, including children	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. China has published protection laws on the minors [7]. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not employ and is not complicit in any form of child labor. According to the Employee list, it could be confirmed that the Project does not employ any child labor.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	An environmental impact assessment (EIA) report of the project was approved by Ordos City Ecological Environment Bureau Ejin Horo Banner Branch on 12-Oct-2023.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified[8]	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	No discrimination occurred. China has published "Labor Law"[9] and "Women's Rights Protection Law", which could prohibit such phenomena. The project abides by the rules of

[7] http://www.gov.cn/xinwen/2020-10/18/content_5552113.htm

[8] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

[9] https://www.gov.cn/banshi/2005-05/25/content_905.htm

		equality and does not involve and is not complicit in any form of discrimination.
Sexual harassment	No risk identified	No sexual harassment had occurred. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena. The program adheres to the principle of respecting women, does not involve and is not complicit in any form of sexual harassment.
Equal pay for equal work	No risk identified	As a project combined with positive environmental, economic, and sustainable development benefits, the project is to generate clean electricity from the LFG sources to replace fossil fuel power. Qualified residents, both men and women, are recruited to work for the project. The project follows the provisions of the "Constitution"[10] and "Labor Law" on equal pay for men and women for equal work. During stakeholders' consultation process, comments were collected from the local people, including both men and women.
Gender equity in labor and work	No risk identified	As a project combined with positive environmental, economic, and sustainable development benefits, the project is to generate clean electricity from the LFG sources to replace fossil fuel power. The project abides by the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in "Women's Rights Protection Law" to provide support and benefits to women. Qualified local residents, both men and women, are recruited to work for the project. The project

[10] https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm

		follows the provisions of the "Constitution"[11] and "Labor Law" on equal pay for men and women for equal work. During stakeholders' consultation process, comments were collected from the local people, including both men and women. The project abides by the rules of equality. Regarding the gender equality, detailed enforcement rules are regulated in "Labor Law" and "Women's Rights Protection Law".
Forced labor	No risk identified	There will be no forced labor on this project. China has promulgated the "Labor Law" to protect the legitimate rights and interests of workers and prohibit forced labor.
Child labor	No risk identified	China has published protection laws on the minors. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not employ and is not complicit in any form of child labor. According to the Employee list, it could be confirmed that the Project does not employ any child labor.
Human trafficking	No risk identified	The project will not use victims of human trafficking. Human trafficking is a serious crime that is strictly regulated and combated in the Chinese legal system. The relevant laws and regulations mainly include the "Criminal Law"[12] and the "Criminal Procedure Law"[13].

[11] https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm

[12] https://flk.npc.gov.cn/detail2.html?ZmY4MDgxODE3OTZhNjM2YTAxNzk4MjJhMTk2NDBjOTI

[13] https://flk.npc.gov.cn/detail2.html?ZmY4MDgwODE2ZjEzNWY0NjAxNmYxZDFiODFiMDEzNTE%3D

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	Human rights protection is formally embodied in the regulations in China. China fully recognizes international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples. China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties–the International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labor Law. This project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in “Women’s Rights Protection Law” to provide support and benefits to women. The project proponent and its activities are strictly regulated by local and national regulations, which obligate the project activities to in line with the international human right law and conventions. The legal approval by authorities demonstrates that the project does not and will not violate the regulations. According to the relevant provisions of the Labor Law and Women’s Rights Protection Law of China, the project continues to recognize, respect, and promote the protection of the rights of IPs, LCs, and customary rights holders.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	According to the EIA report, no indigenous peoples present in or within the area of influence of the Project. According to the EIA report, the Project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The feasibility study report of the project was approved by the Ejin Horo Banner Energy Bureau on 08-Nov-2023. An environmental impact assessment (EIA) report of the project was approved by Ordos City Ecological Environment Bureau Ejin Horo Banner Branch on 12-Oct-2023. The project is located at Ejin Horo Banner landfill site, #4 Community, Zhang Gangtu Village, Altenheige Town, Ejin Horo Banner, Ordos City, Iner Mongolia Autonomous Region, PR China. According to the ecological environment control zoning map of Ordos City, the project is not located in nature reserves, scenic spots, drinking water source protection areas and other areas requiring special protection. After the operation of the project, the prevention measures proposed in the report will not affect the ecological environment and meet the requirements of the red line of ecological protection. Based on questionnaire surveys and stakeholder consultation meetings, residents have expressed support for the project. The project was constructed with the consent of government departments at all levels. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not applicable as no property rights are involved in the project.
Summary of the benefit sharing plan	Not applicable as no property rights are involved in the project.
Approval and dissemination of benefit sharing plan	Not applicable as no property rights are involved in the project.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The project is an LFG recovery and utilization project. According to the EIA report, it does not affect biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	The Project's purpose is to supply electricity from LFG. Therefore, the impact of the Project on soil degradation and soil erosion is negligible.
Water consumption and stress	No risk identified	The Project's purpose is to supply electricity from LFG. Therefore, the impact of the Project on water consumption and stress is negligible.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

Species and habitat	Not applicable as the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.
Areas needed for habitat connectivity	Not applicable as the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	According to EIA, the project isn't adjacent to habitats for rare, threatened, or endangered species.
Areas for habitat connectivity	No risk identified	According to EIA, the project doesn't belong to areas for habitat connectivity.

2.4.2 Introduction of Species

According to EIA, no planting or species introduction is in the project. This section is N/A.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

According to EIA, no invasive species exist in the project area. Therefore, the following table isn't applicable.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	N/A

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No risk identified	No risk identified

2.4.3 Ecosystem Conversion

The project isn't ARR, ALM, WRC or ACoGS project. Hence, not applicable.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	N/A

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.G	Landfill methane recovery	10.0
Methodology	AMS-I.D	Grid connected renewable electricity generation	18.0
TOOL	TOOL04	Emissions from solid waste disposal sites	08.1

TOOL	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
TOOL	TOOL32	Positive lists of technologies	04.0

3.2 Applicability of Methodology

The selected methodology AMS-III.G. (version 10.0) and AMS-I.D. (version 18.0) are appropriate to LFG project activities, where the baseline scenario is the atmospheric release of the LFG, and that all or part of the electricity exported to the grid is the electricity generation in existing and/or new grid-connected power plants. In this case, the LFG is released to atmosphere prior to the implementation of the project, electricity generated by the project is exported to North China Grid Company (NCGC).

The project fulfils the following applicability conditions of the methodology:

Methodology ID	Applicability condition	Justification of compliance
AMS-III.G.	2.This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.	Applicable. The project consists in capturing landfill gas (which contains methane) from the Ejin Horo Banner landfill site, which is used for disposal of residues from human activities.
	3.Different options to utilize the recovered landfill gas as detailed in paragraph 4 for “AMS-III.H. ： Methane recovery in wastewater treatment” (version 19.0) are eligible for use under this methodology. The relevant procedures in AMS-III.H. shall be followed in this regard. Paragraph 4 of AMS-III.H.: The recovery biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energygeneration directly; (b) Thermal or mechanical, electrical energy generation after bottling of	Applicable as (a).

	upgraded biogas, in this case additional guidance provided inthe appendix shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided inthe appendix shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	
	4.Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity.	Applicable
	5.The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity	The implementation of the project does not reduce the amount of organic waste that would be recycled in the absence of the project. All the solid waste is disposed in the Ejin Horo Banner landfill site.
	6.This methodology is not applicable if the management of the solid waste disposal site SWDS) in the project activity is	Not applicable.

	deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement). Such changes may include, for example, the addition of liquids to a SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the SWDS or changing the shape of the SWDS to increase methane production.	
AMS-I.D.	4.This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant (s); (c) Involve a retrofit of (an) existing plant (s); (d) Involve a rehabilitation of (an) existing plant(s)/unit (s) or; (e) Involve a replacement of (an) existing plant(s).	Applicable. The project installs a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project, which corresponds to point (a).
	5. Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligibleto apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power densityof the project activity, as per definitions givenin the project emissions section, is	Not applicable. The project is not a hydro power plant.

	greater than $4W/m^2$; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than $4W/m^2$.	
	6.If the new unit has both renewable and non- renewable components (e.g. a wind/ diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel. the capacity of the entire unit shall not exceed the limit of 15 MW.	Not applicable. The project does not use non-renewable components nor co-fires fossil fuels). Anyway, the total installed capacity is below 15MW.
	7.Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable. The project only involves electricity generation.
	8.In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not applicable. The project does not involve addition of renewable energy generation units at an existing renewable power generation facility.
	9.In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.	Not applicable.
	10.In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation	Applicable. The project is designed to install two engines to combusted LFG of Ejin Horo Banner landfill site, which mainly contains 50% methane.

	for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-I.C. Thermal energy production with or without electricity shall be explored.	
	11.In case biomass Is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass "shall apply.	Not applicable.
TOOL 07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable. This tool is applied to estimate the OM, BM and/or CM.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off- grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid	Applicable. The emission factor for the project electricity system is calculated for grid power plants only.

	power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not applicable. The project electricity system is located in China.
	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	Not applicable.
TOOL04	(a)Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting methane (e.g. "ACM0001: Flaring or use of landfill gas). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM- PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS)	Applicable. The project collects the methane from the Ejin Horo Banner landfill site and uses it to generate the electricity.
	(b)Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW)is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a	Not applicable.

	SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions	
TOOL32	The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM POA for demonstrating their additionality.	Applicable.
	This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Applicable. This tool is applied in conjunction with small-scale methodology AMS-III.G.（version 10.0.）

3.3 Project Boundary

The project boundary is the site of the project activity—Ejin Horo Banner landfill site, where the LFG is captured and used, and includes all the power sources connected to the NCGC, which expands throughout Beijing City, Tianjin City, Hebei Province, Shanxi Province, Shandong Province, Inner Mongolia Autonomous Region.

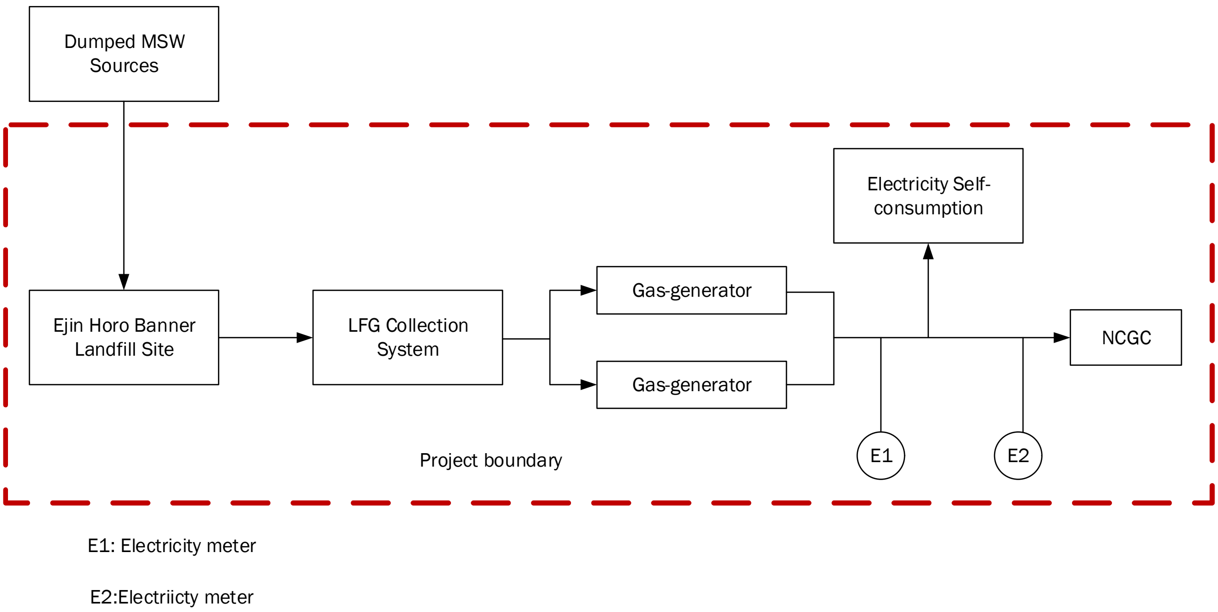


Figure 3.1 Diagram of Project Boundary

Table 3.1 Summary of greenhouse gases and sources included in and excluded from the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of wasteat the SWDS site	CO_2	No	CO_2 emissions from the decomposition of organic waste arenot accounted since the CO_2 is also released under the project activity.
		CH_4	Yes	The major source of emissions in the baseline.
		N_2O	No	N_2O emissions are small comparedto CH_4 emissions from landfills. Exclusion of this gas is conservative.
	Emission from electricity generation	CO_2	Yes	The major source of emissions in the baseline.
		CH_4	No	Excluded for simplification, this is conservative.
		N_2O	No	Excluded for simplification, this is conservative.
	Emissions from heatgeneration	CO_2	No	No heat generation is included in the project.
		CH_4	No	No thermal energy generation is included in the project.
		N_2O	No	No thermal energy generation is included in the project.
	Emissions from the useof natural gas	CO_2	No	Excluded for simplification. This is conservative
		CH_4	No	No supply of LFG is included in the project.
		N_2O	No	Excluded for simplification. This is conservative
Project	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO_2	No	Excluded because there is no fossil fuel consumption.
		CH_4	No	Excluded because there is no fossil fuel consumption.
		N_2O	No	Excluded because there is no fossil fuel consumption.

Source		Gas	Included?	Justification/Explanation
	Emissions from electricity consumption due to the project activity	CO_2	Yes	Main emission source.
		CH_4	No	Excluded for simplification. This emission source is very small compared to CO_2 emissions.
		N_2O	No	Excluded for simplification. This emission source is very small compared to CO_2 emissions.
	Emissions from flaring	CO_2	No	Excluded because there is no flaringin the project.
		CH_4	No	Excluded because there is no flaringin the project
		N_2O	No	Excluded because there is no flaringin the project
	Emissions from distribution of LFG usingtrucks and dedicated pipelines	CO_2	No	No supply of LFG is included in the project.
		CH_4	No	No supply of LFG is included in the project.
		N_2O	No	No supply of LFG is included in the project.

3.4 Baseline Scenario

According to section 5.3 of the methodology AMS-III.G (version 10.0), Section 5.2 of methodology AMS-I.D (Version 18.0), the project chooses simplified procedures to identify the baseline scenario.

The LFG from Ejin Horo Banner landfill site is vented to the atmosphere prior the implementation of the project for safety concerns, and the majority electricity generated by the LFG will be exported to the NCGC, and a little of which is consumed by the project.

LFG from Ejin Horo Banner landfill site is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the NCGC, which is dominated by coal-based power plants.

3.5 Additionality

The project uses the following steps to demonstrate additionality. The project meets the additionality criteria as outlined in the VCS methodology as:

Step 1: Regulatory Surplus. the project is not mandated by any legal requirements. Please refer to section 3.5.1 below for details

Step 2: Positive List. The project conforms to the positive list as stipulated in the applies the simplified procedures in section 5.3 of the methodology AMS-III.G (version 10.0), Section 5.2 of methodology AMS-I.D (Version 18.0) and CDM TOOL32- “Positive lists of technologies " (version 04.0). Please refer to section 3.5.2 below for details.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

According to the relevant applicable laws and regulations at the time of submission of registration,

i.e. Renewable Energy Law of the Peoples Republic of China which came into effect on 01-Jan-2006, Agenda for China’s 14th Five-year Plan (2021-2025), there are no new national and / or sectoral policies that could affect the baseline Scenario during the crediting period. Therefore, the project is surplus to regulatory requirements

3.5.2 Additionality Methods

According to the section 5.3 of the methodology AMS-III.G (version 10.0), section 5.2 of methodology AMS-I.D (Version 18.0) and CDM TOOL32(Version 4.0), project participants may either apply the simplified procedures in section 5.3.1 below or the procedures in section 5.3.2 to select the most plausible baseline scenario and demonstrate additionality. The project chooses simplified procedures to demonstrate additionality. Tool32 (Positive lists of technologies) is used to demonstrate additionality.

Requirements of tool 32 (Positive lists of technologies)	The project

5.1Waste handling and disposal 5.1.1Landfill gas recovery and its gainful use The project activities and PoAs at new or existing landfills (greenfield or brownfield)are deemed automaticallyadditional if it is demonstrated that prior to the implementation of the project activities and PoAs the landfill gas(LFG)was only vented and or flared (in the case of brownfield projects)or would have been only vented and/orflared(in the case of greenfield projects)but not utilized forenergy generation, and that under the project activities andPoAs any of the following conditions are met: (a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW; (b) The LFG is used to generate heat for internal or external consumption; (c)The LFG is flared.	The LFG from Ejin Horo Banner landfill site is vented to the atmosphere prior the implementation of the project for safety concerns. Total nameplate capacity of the project is 2MW, which is below 10MW.

Conclusion: the project is additional.

3.6 Methodology Deviations

No methodology deviations are applied for the project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The project utilizes the LFG for generation to substitute the equivalent electricity supplied by the grid, resulting in CH_4 and CO_2 emissions, which will be calculated as follows according to the methodology AMS-III.G. and AMS-I.D.

Baseline emissions associated with the SWDS

$$BE_{CH4,y} = \eta_{PJ} \times BE_{CH4,SWDS,y} - (1 - OX) \times F_{CH4,BL,y} \times GWP_{CH4}$$ Equation (1)

Where:

$BE_{CH4,y}$	=	Baseline emissions associated with the SWDS in year y (tCO_2e/yr)
η_{PJ}	=	Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50% may be used
$BE_{CH4,SWDS,y}$	=	Methane emission potential of a solid waste disposal site (in tCO_2e)
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering waste) (dimensionless). A default value of 0.1 may be used
$F_{CH4,BL,y}$	=	Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (t CH_4). The relevant procedures in “ACM0001: Flaring or use of landfill gas” may be followed, as well as taking into account the compliance with the relevant local laws and regulation if such laws and regulations exist
GWP_{CH4}	=	Global Warming Potential for methane

Ex ante determination of $BECH_{4,SWDS,y}$

$$BE_{CH4,SWDS,y} = \varphi_y \times (1 - f_y) \times GWP_{CH4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^{y} \sum_{j} (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{k_j})$$

Equation (2)

Where：

$BE_{CH4,SWDS,y}$	=	Baseline methane emissions occurring in the year y
φ_y	=	Model correction factor to account for model uncertainties for year y
f_y	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emission of methane to the atmosphere in year y
GWP_{CH4}	=	Global Warming Potential of methane
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	=	Fraction of the methane in the SWDS gas (volume fraction)

$DOC_{f,y}$	=	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
MCF_y	=	Methane correction factor for year y
$W_{j,x}$	=	Amount of organic waste type j disposed/prevented from disposal in the SWDS in the year y (t)
DOC_j	=	Fraction of degradable organic carbon in the waste type j (weight fraction for year y
k	=	Decay rate for the waste type(1/yr)
j	=	Type of residual waste or types of waste in the MSW
x	=	Years of the crediting period for which waste is disposed of at the SWDS, extending from the first year in the time period(x=1) to year y(x=y)
y	=	Years of the crediting period for which methane emissions are calculated

Determination of $FCH_{4,BL,y}$

This section provides a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odor concerns, or for other reasons (collectively referred to as requirement in this section). The four cases in Table 4.1 are distinguished. The appropriate case should be identified, and the corresponding instructions followed.

Table 4.1. Cases for determining methane captured and destroyed in the baseline

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Currently China has regulations in place to deal with the management of landfills and to encourage utilization of LFG. Those regulations are:

“Standard for Pollution Control on the Landfill Site of Municipal Solid Waste” (GB 16889-2008), which became effective in 2008, issued by the Environment Protection Administration.

“Technical Code for Municipal Solid Waste Sanitary Landfill” （GB 50869-2013), issued by the Ministry of Construction in 2013.

According to item 5.15 of GB16889-2008, if the designed landfill capacity is more than 2.5 milliontons and the landfill thickness is more than 20m, methane utilization facilities or flare burningfacilities shall be built to treat the landfill gas containing methane. For municipal solid waste landfills smaller than the above scale, technologies that can effectively reduce methanegeneration and emission shall be adopted or flare combustion facilities shall be used to treatmethane containing landfill gas.

Item 11.1.1 of GB 50869-2013 stipulates that the landfill site must be equipped with effective landfill gas drainage facilities to prevent the natural accumulation and migration of landfill gas, causing fire and explosion. Item 11.1.3 stipulates that if the landfill does not have the conditions for landfill gas utilization, the flare method shall be adopted for combustion treatment, and the process that can effectively reduce the generation and emission of methane shall be adopted. The old landfills that are not safe and stable should be equipped with effective landfill gas drainage facilities. Among them, item 11.1.1 is mandatory and must be strictly implemented.

In fact, the LFG of Ejin Horo Banner landfill is emitted to atmosphere without LFG capture system prior the implementation of the project. Therefore, Case 2 listed in the table 4.1 is applicable for the project.

The requirements above don't specify any amount or percentage of LFG that should be destroyed.

In this situation:

$$F_{CH4,BL,R,y} = 0.2 \times F_{CH\ \ ,PJ,capt,y} \qquad \text{Equation (4)}$$

Where:

$F_{CH4,BL,R,y}$	=	Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (tCH_4/yr)
$F_{CH4,PJ,capt,y}$	=	Amount of methane in the LFG which is captured in the project activity in year y (tCH_4/yr)

Baseline emissions associated with electricity generation ($BE_{EC,\ y}$)

$$BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y} \qquad \text{Equation (5)}$$

Where:

$BE_{EC,y}$	=	Baseline emissions associated with electricity generation in year y (tCO_2)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y(MWh)

$EF_{grid,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO_2/MWh)

Calculation of $EG_{PJ,y}$

The project is the installation of a greenfield power plant, therefore:

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad \text{Equation (6)}$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y(MWh)

According to AMS I.D, the emission factor shall be calculated in a transparent and conservative manner as follows:

(a) A combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the "Tool to calculate the emission factor for an electricity system"; or

(b) The weighted average emissions (in tCO_2/MWh) of the current generation mix. The data of the year in which project generation occurs must be used.

According to "Tool to calculate the emission factor for an electricity system" (version 07.0). The following six steps are applied to determine OM, BM, and CM used for calculating the project baseline emissions:

Step 1: Identify the relevant electricity systems;

Step 2: Choose whether to include off-grid power plants in the project

Step 3: Select a method to determine the operating margin (OM);

Step 4: Calculate the operating margin emission factor according to the

Step 5: Calculate the build margin (BM) emission factor;

Step 6: Calculate the combined margin (CM) emission factor.

Step 1: Identity the relevant electricity systems

According to the Tool to calculate the emission factor for an electricity system (Version 07.0), project participants may delineate the project electricity system using any of the following options:

Option 1. A delineation of the project electricity system and connected electricity systems published by the DNA or the group of the DNAs of the host country(ies), In case a delineation isprovided by a group of DNAs, the same delineation should be used by all the project participantsapplying the tool in these countries.

Option 2. A delineation of the project electricity system defined by the dispatch area of thedispatch center responsible for scheduling and dispatching electricity generated by the projectactivity. Where the dispatch area is controlled by more than one dispatch center, i.e. layereddispatch area, the higher-level area shall be used as a delineation of the project electricity system(e.g. where regional dispatch centers are required to comply with dispatch orders of the nationaldispatch center then area controlled by the national dispatch center shall be used);

Option 3. A delineation of the project electricity system defined by more than one independentdispatch areas, e.g. multi-national power pools.

The Chinese DNA has published a delineation of the project electricity system and connectedelectricity systems, Option 1 is applied for the project. According to the delineations, the NCGC is identified as the relevant electric power system of the project, which includes Beijing City, Tianjin City, Hebei Province, Shanxi Province, Shandong Province, Iner Mongolia Autonomous Region Power Grids.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional). Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Following the calculation of the Ministry of Ecology and Environment of the People's Republic of

China, and statistical data is available, Option I is chosen.

Step 3. Select a method to determine the operating margin (OM)

"Tool to calculate the emission factor for an electricity system (Version 7.0)" offers four methods

for the calculation of the operating margin emission factor(s) ($EF_{grid,OM,y}$):

(a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

Method (a) -Simple OM is chosen for calculation and low-cost/must-run resources constitute less than 50% of the total grid generation in average of the five most recent years[14].

For simple OM, the emission factor can be calculated using either of the two following data vintages:

[14] 2023 China baseline emission factor of regional power grid.

(a) Ex ante option: If the ex-ante option is chosen, the emission factor is determined once at thevalidation stage, thus no monitoring and recalculation of the mission factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average,based on the most recent data available at the time of submission of the CDM-PDD to theDOE for validation. For off-grid power plants, use a single calendar year within the 5 mostrecent calendar years prior to the time of submission of the CDM-PDD for validation.

(b) Ex post option: If the ex-post option is chosen, the emission factor is determined for the yearin which the project activity displaces grid electricity, requiring the emissions factor to beupdated annually during monitoring. If the data required to calculate the emission factor foryear y is usually only available later than six months after the end of year y, alternatively theemission factor of the previous year y-1 may be used. If the data is usually only available 18months after the end of year y, the emission factor of the year proceeding the previous year y-2 may be used. The same data vintage (y, y-1 or y-2) should be used throughout all creditingperiods

Project participant employs ex ante option for its operation margin calculation.

Step 4. Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated by one of the following two options:

(a) Option A: Based on the net electricity generation and a CO_2 emission factor of each power unit; or

(b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if:

(i) The necessary data for Option A is not available: and

(ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources, and the quantity of electricity supplied to the grid by these sources is known; and

(iii) Off-grid power plants are not included in the calculation (i.e., if Option I have been chosen in Step 2).

Since the data of each power plant/unit is unavailable, Option A is not applicable to the project. The project adopts Option B to calculate the operating margin emission factor ($EF_{grid,OM,y}$) of NCGC.

$$EF_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_y} \quad \text{Equation (7)}$$

Where：

$EF_{grid,OMsimple,y}$ = Simple operating margin CO_2 emission factor in year y (tCO_2/MWh)

$FC_{i,y}$ = Amount of fuel type i consumed in the project electricity system inyear y (mass or volume unit)

$NCV_{i,y}$ = Net calorific value (energy content) of fuel type i in year y (GJ/massor volume unit)

$EF_{CO2,i,y}$ = CO_2 emission factor of fossil fuel type i in year y (tCO_2e/GJ), 2006 IPCC Guidelines for default values

EG_y = Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must run power plants/units in year y (MWh)

i = All fuel types combusted in power sources in the project electricity system in year y

y = The relevant year as per the data vintage chosen in Step 3

For North China Grid Company and Southern Power Grid, use the adjusted simple OM method to calculate their OM emission factor. The adjusted simple OM divides the generator sets into two categories: low cost/must run power plants and all the other power plants except low cost/ must run power plants. According to the weighted average of each unit's annual power supply emission factor, the probability weights of the two units participating in generation scheduling (λ and 1 -λ) are obtained. Finally, the unit power supply emission factor of the two types of generating units is adjusted according to the participation in generation. The probability weights of degrees are weighted and averaged to obtain the adjusted simple OM. The specific calculation formula is as follows:

$$EF_{grid,OM-adj,y} = \left(1-\lambda_y\right) \times EF_{grid,OMsimple,y} + \lambda_y \times \frac{\sum_k EG_{k,y} \times EF_{EL,k,y}}{\sum_k EG_{k,y}} \qquad \text{Equation (8)}$$

Where:

$EF_{grid,OMsimple,y}$ = Simple operating margin CO_2 emission factor in year y (tCO_2/MWh), according to the equation (6)

$EG_{k,y}$ = Net electricity generated and delivered to the grid by low-cost/must run power plants/units type k in year y(MWh)

$EF_{EL,k,y}$ = CO_2 emission factor by low-cost/must run power plants/units type k in year y (tCO_2/MWh)

k = Type of low-cost/must run power plants/units in year y

λ_y	=	Probability weight of low-cost/must run power plants/units participating in generation scheduling in year y
y	=	The relevant year as per the data vintage chosen in Step 3

If available, $NCV_{i,y}$ and EFCO2,i,yfrom the fuel supplier of the power plants in invoices may be used; or regional or national average default values may be used. In this PD, $NCV_{i,y}$ of different fuels are obtained from China Energy Statistical Yearbook 2018. Emission factors ($EF_{CO2,i,y}$) of each type of fossil fuel come from 2019 refinement to 2006 IPCC default values.

The Simple OM Emission Factor ($EF_{grid,OMsimple,y}$) of the project is calculated based on the fuel consumption data for electricity generation of the China Southern Power Grid, not including those of low-operating cost and must-run power plants, such as wind power, hydropower and nuclear etc. These data are obtained from the China Electric Power Yearbook (2020~2022, published annually) and China Energy Statistical Yearbook (2017~2019). Based on these data, the Simple OM Emission Factor ($EF_{grid,OMsimple,y}$) of the North China Grid is calculated as 0.9350 tCO_2/MWh. Details of the calculations and the published data from the Chinese DNA[15], which uses official national statistics.

Step 5. Calculate the build margin emission factor

In terms of vintage of data, project participants can choose between one of the following twooptions:

a) Option 1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

b) Option 2: For the first crediting period, the build margin emission factor shall be updated annually, ex-post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The project applies option 1 to calculate the build margin emission factor ex-ante.

[15] https://ccer.cets.org.cn/notice/noticeDetail?bulletinInfoId=1175122354980917248

According to the “Tool to calculate the emission factor for an electricity system”, the sample groupof power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\ units}$) and determine their annualelectricity generation ($AEG_{SET\text{-}5\text{-}units}$, in MWh);

(b) Determine the annual electricity generation of the project electricity system, excluding power units registered as CDM project activities (AEG_{total}, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEGtotal (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET\text{-}\geq 20\%}$, in MWh);

(c) From $SET_{5\text{-}units}$ and $SET_{\geq 20}$ percent select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SETsample started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. Ignore steps (d), (e) and (f).

Otherwise:

(d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activity, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample\text{-}CDM}$) the annual electricity generation ($AEG_{SET\text{-}sample\text{-}CDM}$, in MWh);

If the annual electricity generation of that set is comprises at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET\text{-}sample\text{-}CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample\text{-}CDM}$ to calculate the build margin. Ignore steps (e) and (f).

(e) Include in the sample group $SET_{sample\text{-}CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sampleCDM}$>10yrs)

The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$$ Equation (9)

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (t CO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

Under the current circumstances in China, the power plants consider the Build Margin data as important business data and will not have them published. Therefore, it is difficult to obtain the data of five power plants that have been put into operation most recently or the newly installed power plant capacity additions in the electricity system that comprise 20% of the system generation.

According to the instructions of China DNA, for the determination of the set of samples, a sample merging processing in some degree has been adopted due to that the power generation data, energy consumption data or thermal efficiency data of each plant cannot be consulted in the public statistical data. In this calculation, the newly installed power units in the past years are classified by year, province and power generation technology, and the same type of newly installed power units in the same province and in the same year are bundled as a "newly installed power units".

The power generation of each "newly installed power units" in the most recent year y is estimated based on its installed capacity and the number of power generation utilization hours in year y. The formula is as follows:

$$EG_{m,y} = CAP_m \times H_{m,y}$$ Equation (10)

Where:

$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

CAP_m	=	Installed capacity of electricity generated and delivered to the grid by power unit m in year y (MW)
$H_{m,y}$	=	The number of power utilization hours (h) of electricity generated and delivered to the grid by power unit m in year
y	=	And it selects the average utilization hours of similar units in the province in which it is in year y;
m	=	The sample group of power units

The power unit m is selected from the "newly installed power plants" in the most recent year y (For the calculation of the grid BM in 2023, the y is equal to 2021) to the "newly installed power plants" in the earlier year, until the cumulative power generation reaches 20% of the total power generation in the year y.

Since the newly installed power units of the same type (k) in the same province (A) and the same year (t) are bundled into the “newly installed power units”, the CAP_m is equal to the statistical data of recent installed capacity of a given unit type(k) each year(t) in a given province (A).

$$CAP_m = CAP_{m|m=(A,t,k)} = CAP_{A,t,k} \quad \text{Equation (11)}$$

Where:

CAP_m	=	Installed capacity of electricity generated and delivered to the grid by power unit m in year y (MW), and m is equivalent to an established combination of (A, t, k)
$CAP_{A,t,k}$	=	Capacity of newly installed power units of a given province (A), given year (t), and given unit type (k) (MW)
A	=	It is the various provincial regions covered by the regional power grid
t	=	It is the sampling year of the "newly installed power units". For the calculation of the grid BM in 2023, t is equal to 2021,2020，2018, until the units that comprise at least 20 percent of the system generation in 2021
k	=	It is the power generation technology classification of ‘newly-installed power units’, which is divided into: hydro power, coal-thermal power, gas-

thermal power, oil-thermal power, Waste-thermal power plant, other thermal power[16] nuclear power, wind power, and others[17]

As per tool, the CO_2 emission factor of each power unit m ($EF_{EL,m,y}$) should be determined as per the tool in Step 4 section 6.4.1 for the simple OM, using Options A1, A2 or A3, using for y the most recent historical year for which electricity generation data is available, and using form the power units included in the build margin.

Because current statistics data cannot separate each power plant, for a power unit m, only data on electricity generation and the fuel types used is available. So, the option A2 is selected, the emission factor should be determined based on the CO_2 emission factor of the fuel type used and the efficiency of the power unit, as follows:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \quad \text{Equation (12)}$$

Where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
$EF_{CO2,m,i,y}$	=	Average CO_2 emission factor of fuel type i used in power unit m in year y (tCO_2/GJ)
$\eta_{m,y}$	=	Average net energy conversion efficiency of power unit m in year y (dimensionless)
3.6	=	Conversion coefficient of thermal work equivalent of electricity (GJ/MWh)

According to Equation (11), the unit electricity emission factor of the hydro-power, nuclear power, wind power, solar power, other thermal power, and others power generation technology in the "newly-installed power units" samples are is zero. The emission factor per unit of electricity for power generation from coal, gas, oil and electricity waste power is calculated based on Equation (11). Since the average net energy conversion efficiency of each sample ($\eta_{m,y}$) cannot be obtained, the power supply thermal efficiency of the best commercialized technology of coal, gas, oil and waste power ($\eta_{Best,m,y}$) is better than $\eta_{m,y}$. It is conservative to use $\eta_{Best,m,y}$ for the calculation of $EF_{EL,m,y}$. Therefore, the formula is as follows:

$$EF_{Best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{Best,m,y}} \quad \text{Equation (13)}$$

Where:

[16] Refer to waste heat and pressure, straw, bagasse, forest wood power generation

[17] Refer to power generation such as geothermal energy and ocean energy

$EF_{Best,m,y}$	=	Emission factor of power unit m with the best technology commercially available in year y (tCO_2/MWh)
$\eta_{Best,m,y}$	=	Power generation efficiency of the best technology commercially available in year y (dimensionless)
m	=	Sample of "newly installed power units" generated by coal, gas, oil and waste incineration in a province in a certain year

The emission factors of coal, natural gas, fuel oil and municipal solid waste (MSW) are sourced from 2006 IPCC Guidelines for National Greenhouse Gas Inventories, and the lower values of the 95% confidence interval are applied to be conservative.

No fuel oil power units were constructed in 2021. The power generation efficiency of the best technology commercially available for fuel oil units comes from Baseline Emission Factors for Regional Grids in China of previous years, i.e., 52.9%.

According to the above steps, the build margin emission factor, $EF_{grid,BM,y}$ of the NCGC is calculated to be 0.3020 tCO_2/MWh, which is confirmed by 2023 Baseline Emission Factors for Regional Power Grids in China published by Ministry of Ecology and Environment of the People's Republic of China[18].

Step 6. Calculate the combined margin emission factor

Combined Margin emission factor($EF_{grid,CM,y}$) is calculated as the weighted average of the operating margin emission factor($EF_{grid,OM,y}$) and the build margin emission factor ($EF_{grid,BM,y}$),where theweights ω_{OM} and ω_{BM} ,by default, are 0.5 and 0.5 in the crediting period, and $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ are calculated as described above and are expressed in tCO_2/MWh.

$$EF_{grid,CM,y} = \omega_{OM} \times EF_{grid,OM.y} + \omega_{BM} \times EF_{grid,BM,y} \quad \text{Equation (14)}$$

$$EF_{grid,CM,y} = 0.5 \times 0.9350 + 0.5 \times 0.3020 = 0.6185$$

The $EF_{OM,y}$, $EF_{BM,y}$ and $EF_{grid,CM,y}$ are ex-ante calculation and are fixed during the credit period.

Table 4.2: Necessary values to calculate the baseline emissions associated with the SWDS

Parameter	Value	Data source
OX	0.1	Using IPCC 2006 Guidelines default value

[18] https://ccer.cets.org.cn/notice/noticeDetail?bulletinInfoId=1175122354980917248

GWP_{CH4}	28	IPCC fifth Assessment Report(AR5)
η_{PJ}	80%	FSR of the project
φ_y	0.75	Emissions from solidwaste disposal sites"(version 08.1)
OX_{top_layer}	0.1	Emissions from solidwaste disposal sites"(version 08.1)
f_y	0	"Emissions from solid waste disposal sites" (version 08.1)
F	0.5	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
$DOC_{f,y}$	0.5	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
MCF_y	1.0	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
$W_{j,x}$	Annual landfill MSW (tons) Year / Annual landfill MSW (tons): 2010 / 43,655 2011 / 69,558	

2012	49,935
2013	79,565
2014	85,097
2015	91,013
2016	97,340
2017	103,553
2018	110,163
2019	116,574
2020	122,710
2021	128,492
2022	133,846
2023	138,700
2024	140,087
2025	141,488
2026	142,903

Organic waste type j fraction on wet basis:

Waste type j	Weight Fraction (wet waste)
W_1-Wood and wood products	0.107
W_2-Pulp, paper and cardboard	0.107
W_3-Food, food waste, beverages and tobacco	0.586
W_4-Textiles	0.009

<table>
<tr><td></td><td><table>
<tr><td>W_5-Garden, yard and park waste</td><td>0.005</td></tr>
<tr><td>W_6-Glass, plastic, metal other inert</td><td>0.186</td></tr>
</table></td><td></td></tr>
<tr><td>DOC_j</td><td><table>
<tr><td>Waste type j</td><td>DOC_j (% wet waste)</td></tr>
<tr><td>Wood and wood products</td><td>43</td></tr>
<tr><td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr>
<tr><td>Food Waste</td><td>15</td></tr>
<tr><td>Textiles</td><td>24</td></tr>
<tr><td>Garden,yard and parkwaste</td><td>20</td></tr>
<tr><td>Glass, plastic, metal, otherinert waste</td><td>0</td></tr>
</table></td><td></td></tr>
<tr><td>k_j</td><td><table>
<tr><td colspan="2">Waste type j</td><td>k_j(MAT≤ 20℃, MAP/PET ＜1)</td></tr>
<tr><td rowspan="2">Slowly degrading</td><td>Pulp, paper, cardboard (other than sludge)</td><td>0.04</td></tr>
<tr><td>Wood, wood products and straw</td><td>0.02</td></tr>
<tr><td>Moderately degrading</td><td>Other(non-food) organic putrescible garden and park waste</td><td>0.05</td></tr>
<tr><td>Rapidly degrading</td><td>Food, food waste, sewage sludge,</td><td>0.06</td></tr>
</table></td><td>“Emissions from solid waste disposal sites” (version 08.1)</td></tr>
</table>

			beverages and tobacco		

Table 4.3 The estimation of baseline emissions associated with the SWDS

Year	η_{PJ}	$BE_{CH4,SWDS,y}$	GWP_{CH4}	$OX_{top,layer}$	$F_{CH4,BL,y}$	$BE_{CH4,y}$
10-Dec-2023 to 31-Dec-2023	80%	2,543.07	28	0.1	509	2,085
01-Jan-2024 to 31-Dec-2024	80%	43,325	28	0.1	8,665	35,526
01-Jan-2025 to 31-Dec-2025	80%	49,056	28	0.1	9,811	40,226
01-Jan-2026 to 31-Dec-2026	80%	50,216	28	0.1	10,043	41,176
01-Jan-2027 to 31-Dec-2027	80%	49,803	28	0.1	9,961	40,838
01-Jan-2028 to 31-Dec-2028	80%	46,640	28	0.1	9,328	38,245

Table 4.4 The estimation of baseline emissions associated with electricity generatio n

Year	$EC_{BL,k,y}$	$EF_{grid,CM,y}$	$BE_{EC,y}$
10-Dec-2023 to 31-Dec-2023	400.86	0.6185	247
01-Jan-2024 to 31-Dec-2024	6,829	0.6185	4,223
01-Jan-2025 to 31-Dec-2025	6,904	0.6185	4,270
01-Jan-2026 to 31-Dec-2026	7,067	0.6185	4,371
01-Jan-2027 to 31-Dec-2027	7,009	0.6185	4,335
01-Jan-2028 to 31-Dec-2028	6,564	0.6185	4,059

4.2 Project Emissions

As per AMS I.D (version 18.0), project emission of the project is zero.

As per AMS III.G (version 10.0), project emission is calculated as per equation below:

$$PE_y = PE_{power,y} + PE_{flare,y} + PE_{process,y}$$ Equation (15)

Where:

PE_y = Project emission in year y (t CO_2e)

$PE_{power,y}$ = Emission from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO_2e)

$PE_{flare,y}$ = Emissions from flaring or combustion of the landfill gas stream in the year y (t CO_2e)

$PE_{process,y}$ = Emission from the landfill gas upgrading process in the year y (t CO_2e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

The project does not involve the emissions from consumption of fossil fuels, only emission fromthe use of electricity for operation of the installed facilities is involved. The project will consumeelectricity delivered from the grid when the project is not in operation. And this part of electricityhas already been deducted from quantity of electricity supplied by the project plant/unit to thegrid when calculating the quantity of the net electricity generation supplied by the projectplant/unit to the grid, which is then used to calculate the baseline emission associated withelectricity generation($BE_{EC,y}$), hence emissions from the use of electricity for the operation of theinstalled facilities is excluded, which means emission form the used of fossil fuel and electricityfor the operation of the installed facilities equal to 0.

The project dosed not claim emission reductions from flaring, therefore, $PE_{flare,y}$ is equal to 0.

The project captures LFG for power generation and isn't involved upgrading process, therefore,$PE_{process,y}$ is equal to 0.

In conclusion, the project emission of the project activity is 0.

4.3 Leakage Emissions

There is no equipment transferred in the project, no leakage effects need to be accounted under AMS-III.G. and AMS-I.D.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

According to AMS-III.G., the emission reduction achieved by the project activity can be estimatedex-ante in the PD by:

$$ER_{y,estimated} = BE_y - PE_y - LE_y \quad \text{Equation (16)}$$

By combining all the above equations, the ex-ante estimate of the emission reduction is:

$$ER_{y,estimated} = \eta_{PJ} \times BE_{CH4,SWDS,y} - (1 - OX) \times F_{CH4,BL,y} \times GWP_{CH4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad \text{Equation（17）}$$

Emission reductions(ex-post)

According to AMS-III.G., The actual emission reduction achieved by the project activity during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times \left(F_{CH4,PJ,y} - F_{CH4,BL,y}\right) \times GWP_{CH} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad \text{Equation (18)}$$

Where:

$F_{CH4,PJ,y}$	=	Methane captured and destroyed/gainfully used by the project activity in the year y (t CH_4)

The project utilizes recovered methane for power generation. Therefore $F_{CH4,PJ,y}$ is calculated as follows, based on the amount of monitored electricity generation without monitoring methane flow and concentration:

$$F_{CH4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4}$$ Equation (19)

Where:

EG_y	=	Electricity generation in year y(MWh)
3600	=	Conversion factor (1 MWh=3600 MJ)
NCV_{CH4}	=	NCV of methane (MJ/NM^3) use default value:35.9 MJ/NM^3
EE_y	=	Energy Conversion Efficiency of the project equipment determined from one of the following options: • Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel if the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation • Default efficiency of 40 per cent

Therefore:

$$ER_{y,calculated} = (1 - OX) \times \left(\frac{EG_y \times 3600}{NCV_{CH4}} \times D_{CH4} - F_{CH4,BL,y}\right) \times GWP_{CH4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y$$ Equation (20)

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
10-Dec-2023 to 31-Dec-2023	2,085	0	0	2,332	0	2,332
01-Jan-2024 to 31-Dec-2024	35,526	0	0	39,749	0	39,749
01-Jan-2025 to 31-Dec-2025	40,226	0	0	44,496	0	44,496
01-Jan-2026 to 31-Dec-2026	41,176	0	0	45,547	0	45,547
01-Jan-2027 to 31-Dec-2027	40,838	0	0	45,173	0	45,173

01-Jan-2028 to 31-Dec-2028	38,245	0	0	42,304	0	42,304
Total	196,011	0	0	217,269	0	217,269

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	OX
Data unit	Dimensionless
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	Based on an extensive review of published literature on this subject, including the IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	Using IPCC 2006 Guidelines default value
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	GWP_{CH4}
Data unit	$tCO_2e/t\ CH_4$
Description	Global warming potential of CH_4
Source of data	IPCC Fifth Assessment Report (AR5)
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value of 28 from IPCC Fifth Assessment Report (AR5). Shall be updated according to any future COP/CMP decisions

Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	η_{PJ}
Data unit	Dimensionless
Description	Efficiency of the LFG capture system that will be installed in the project activity
Source of data	FSR of the project
Value applied	80%
Justification of choice of data or description of measurement methods and procedures applied	Using estimated value in FSR of the project
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	φ_y
Data unit	-
Description	The model correction factor to account for model uncertainties
Source of data	Default value of the tool “Emissions from solid waste disposalsites” (version 08.1)
Value applied	0.75
Justification of choice of data or description of measurement methods and procedures applied	Application A is used to decide the value. And the value of wet conditions for used for the project since the(MAT≤20˚C, MAP/PET＜1).
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	f_y

Data unit	-
Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
Source of data	The tool "Emissions from solid waste disposal sites" (version 08.1)
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	It has been considered in the section 4.1, equation (1). Therefore, f_y=0
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Using IPCC 2006 Guidelines default value
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$DOC_{f,y}$
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposed in the SWDS

Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Using IPCC 2006 Guidelines default value
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	MCF_y
Data unit	-
Description	Methane correction factor
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	The project using application A to decide the value of MCF_y. The Ejin Horo Banner landfill site has controlled placement of waste, mechanicalcompacting and levelling of the waste, So MCF_y value of 1.0 foranaerobic managed solid waste disposal sites should be appliedin line with" Emissions from solid waste disposal sites" (version08.1)
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$W_{j,x}$
Data unit	t
Description	Amount of solid waste type j disposed or prevented from disposal in the SWDS in year x
Source of data	Feasibility study report (FSR) of the project
Value applied	Annual landfilled MSW (tons) Year | Annual landfilled MSW (tons)

2010	43,655
2011	69,558
2012	49,935
2013	79,565
2014	85,097
2015	91,013
2016	97,340
2017	103,553
2018	110,163
2019	116,574
2020	122,710
2021	128,492
2022	133,846
2023	138,700
2024	140,087
2025	141,488
2026	142,903

Organic waste type j fraction on wet basis:

Waste type j	Weight Fraction (wet waste)
W_1-Wood and wood products	0.1070
W_2-Pulp, paper and cardboard	0.1070
W_3-Food, food waste, beverages and tobacco	0.5860
W_4-Textiles	0.0090
W_5-Garden, yard and park waste	0.0050

	W_6-Glass, plastic, metal other inert	0.1860
Justification of choice of data or description of measurement methods and procedures applied	It is calculated as total waste amount dumped in the landfill site in the year x multiplied by organic waste type j fraction on wet basis. Both total waste amount and waste type j fraction are referred to FSR.	
Purpose of data	• Calculation of baseline emissions	
Comments	-	

<table>
<tr><td>Data / Parameter</td><td colspan="2">DOC_j</td></tr>
<tr><td>Data unit</td><td colspan="2">-</td></tr>
<tr><td>Description</td><td colspan="2">Fraction of degradable organic carbon in the waste type j (weight fraction)</td></tr>
<tr><td>Source of data</td><td colspan="2">“Emissions from solid waste disposal sites” (version 08.1)</td></tr>
<tr><td rowspan="7">Value applied</td><td>Waste type j</td><td>DOC_j (% wet waste)</td></tr>
<tr><td>Wood and wood products</td><td>43</td></tr>
<tr><td>Pulp, paper and cardboard (otherthan sludge)</td><td>40</td></tr>
<tr><td>Textiles</td><td>15</td></tr>
<tr><td>Garden, yard and park waste</td><td>24</td></tr>
<tr><td>Glass, plastic, metal, other inertwaste</td><td>20</td></tr>
<tr><td>Wood and wood products</td><td>0</td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td colspan="2">The climate condition[19] of location of Ejin Horo Banner landfill: Average annual temperature:6.2˚C
Mean annual precipitation:358.2mm Potential evapotranspiration:2,563mm</td></tr>
<tr><td>Purpose of data</td><td colspan="2">Calculation of baseline emissions</td></tr>
<tr><td>Comments</td><td colspan="2">-</td></tr>
</table>

[19] http://ordos.nmgqq.com.cn/sqgl/shengtaihj/2021-04-10/6079.html

<table>
<tr><td>Data / Parameter</td><td>k_j</td></tr>
<tr><td>Data unit</td><td>1/yr</td></tr>
<tr><td>Description</td><td>Decay rate for the waste type j</td></tr>
<tr><td>Source of data</td><td>“Emissions from solid waste disposal sites” (version 08.1)</td></tr>
<tr><td>Value applied</td><td>
<table>
<tr><td colspan="2">Waste type j</td><td>k_j(MAT≤20℃, MAP/PET＜1)</td></tr>
<tr><td rowspan="2">Slowly degrading</td><td>Pulp, paper, cardboard (other than sludge), textiles</td><td>0.04</td></tr>
<tr><td>Wood, wood products and straw</td><td>0.02</td></tr>
<tr><td>Moderately degrading</td><td>Other(non-food) organic putrescible garden and park waste</td><td>0.05</td></tr>
<tr><td>Rapidlydegrading</td><td>Food, food waste, sewage sludge, beverages and tobacco</td><td>0.06</td></tr>
</table>
</td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>The climate condition of location of Ejin Horo Banner landfill: Average annual temperature:6.2˚C
Mean annual precipitation:358.2mm Potential evapotranspiration:2,563mm</td></tr>
<tr><td>Purpose of data</td><td>Calculation of baseline emissions</td></tr>
<tr><td>Comments</td><td>-</td></tr>
</table>

Data / Parameter	D_{CH4}
Data unit	t/m³
Description	Density of methane at the temperature and pressure of the landfillgas in the year y
Source of data	Default value as per the latest version of “Project emissions from flaring”

Value applied	0.0007156 at normal conditions
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2/MWh
Description	Operation margin emission factor of NCGC
Source of data	2023 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.9350
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Energy Statistic Yearbook (2020-2022), System of Statistical Survey of Energy Consumption in Public Institutions (National Government Offices Administration, approved by the National Bureau of Statistics, August 2019), China Electric Power Yearbook (2020-2022), Power Industry Statistical Data Collection (2019-2021) and IPCC data.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO_2/MWh
Description	Build margin emission factor of NCGC
Source of data	2023 Bulletin on the Baseline Emission Factors of the ChinaGrids
Value applied	0.3020
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Energy Statistic Yearbook (2020-2022), System of Statistical Survey of Energy Consumption in Public Institutions (National Government Offices

	Administration, approved by the National Bureau of Statistics, August 2019), China Electric Power Yearbook (2020-2022), Power Industry Statistical Data Collection (2019-2021) and IPCC data.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Combine margin emission factor of NCGC
Source of data	Calculated based on "Tool to calculate the emission factor for an electricity system (version 07.0)"
Value applied	0.6185
Justification of choice of data or description of measurement methods and procedures applied	The value is calculated based on $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ as per "Tool to calculate the emission factor for an electricity system (version 07.0)"
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	NCV_{CH4}
Data unit	MJ/Nm^3
Description	Net calorific value of methane at reference conditions
Source of data	AMS-III.G Landfill methane recovery (version 10.0)
Value applied	35.9
Justification of choice of data or description of measurement methods and procedures applied	Default value as described in AMS-III.G
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	EE_y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	Default value of AMS.III.G (Version 10.0)
Value applied	40%(default value)
Justification of choice of data or description of measurement methods and procedures applied	As per paragraph 21 of AMS.III.G (Version 10.0), there are two options to decide the value of EE_y. Option 1: Specification provided by the equipment manufacturer. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for biogas. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation. Option 2: Default efficiency of 40 percent. Default value 40% is applied in the project.
Purpose of data	Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$F_{CH4,BL,y}$
Data unit	tCH_4/yr
Description	Amount of methane in the LFG which is flared due to a requirement in year y
Source of data	Information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odor concerns.
Description of measurement methods and procedures to be applied	According to AMS III.G, the relevant procedures in ACM0001 may be followed to determine parameter $F_{CH4,BL,y}$. Case 2 of section 5.4.1.3 of ACM0001"Flaring or use of landfill gas" (version 19.0) is applied for the project.
Frequency of monitoring/recording	Annually
Value applied	The data below are calculated ex ante:

	<table><tr><td>Year</td><td>Amount of methane in the LFG which is flared due to a requirement (tCH_4)</td></tr><tr><td>10-Dec-2023 to 31-Dec-2023</td><td>508.61</td></tr><tr><td>01-Jan-2024 to 31-Dec-2024</td><td>8,665</td></tr><tr><td>01-Jan-2025 to 31-Dec-2025</td><td>9,811</td></tr><tr><td>01-Jan-2026 to 31-Dec-2026</td><td>10,043</td></tr><tr><td>01-Jan-2027 to 31-Dec-2027</td><td>9,961</td></tr><tr><td>01-Jan-2028 to 31-Dec-2028</td><td>9,328</td></tr></table>
Monitoring equipment	-
QA/QC procedures to be applied	Yearly updated in line with the relevant local laws and regulation if such laws and regulations exist.
Purpose of data	Calculation of baseline emissions
Calculation method	The data is calculated as methane capture and destroyed multiplied by fraction of LFG that is required to be flared due to a requirement.
Comments	Applicable to Case 2 of section 5.4.1.3 of ACM0001"Flaring oruse of landfill gas" (version 19.0)

Data / Parameter	$\rho_{reg,y}$
Data unit	Dimensionless
Description	Fraction of LFG that is required to be flared due to a requirement in year y
Source of data	Information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odor concerns.
Description of measurement methods and procedures to be applied	-

Frequency of monitoring/recording	Annually
Value applied	20%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	Applicable to Case 2 of section 5.4.1.3 of ACM0001"Flaring oruse of landfill gas" (version 19.0)

Data / Parameter	$EG_{PJ,facility,y}$
Data unit	MWh
Description	Quantity of net electricity generated supplied by the project plant/unit in year y
Source of data	Electricity meter E2
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter (bi-directional) installed at the project site. All data will be monitored and archived electronically. Double check by receipt of electricity sales.
Frequency of monitoring/recording	The recording frequency will be hourly measured and record andmonthly aggregated.
Value applied	The data below are calculated ex ante

Year	Quality of net electricity generated supplied by the project plant (MWh)
10-Dec-2023 to 31-Dec-2023	400.86
01-Jan-2024 to 31-Dec-2024	6,829
01-Jan-2025 to 31-Dec-2025	6,904

	<table><tr><td>01-Jan-2026 to 31-Dec-2026</td><td>7,067</td></tr><tr><td>01-Jan-2027 to 31-Dec-2027</td><td>7,009</td></tr><tr><td>01-Jan-2028 to 31-Dec-2028</td><td>6,564</td></tr></table>
Monitoring equipment	Electricity meter E2 is shown in the monitoring system as below.
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	EG_y
Data unit	MWh
Description	Electricity generation in year y
Source of data	Electricity meter E1
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter installed at the project site. All data will be monitored and archived electronically.
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.
Value applied	The data below are calculated ex ante <table><tr><td>Year</td><td>Electricity generated in year y (MWh)</td></tr><tr><td>10-Dec-2023 to 31-Dec-2023</td><td>426.45</td></tr><tr><td>01-Jan-2024 to 31-Dec-2024</td><td>6,829</td></tr><tr><td>01-Jan-2025 to 31-Dec-2025</td><td>6,904</td></tr><tr><td>01-Jan-2026 to 31-Dec-2026</td><td>7,067</td></tr><tr><td>01-Jan-2027 to 31-Dec-2027</td><td>7,009</td></tr></table>

	01-Jan-2028 to 31-Dec-2028 | 6,564
Monitoring equipment	Electricity meter E1 shown in the monitoring system as below.
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	$D_{CH4,y}$
Data unit	t/m^3
Description	Density of methane at the temperature and pressure of the landfill gas in year y ($tons/m^3$). If $LFG_{i,y}$ is reported at normal conditionsof temperature and pressure, the density of methane is also determined at normal conditions
Source of data	Tool 08 "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (version 03.0)
Description of measurement methods and procedures to be applied	The project employs landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas is displayed. According to the AMS-III.G., the density of methane is also determined at normal conditions
Frequency of monitoring/recording	-
Value applied	0.0007156
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions

Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the landfill gas
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	℃
Description	Temperature of the landfill gas
Source of data	-
Description of measurement methods and procedures to be applied	The project employs landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas is displayed. According to the AMS-III.G., there is no need for separate monitoring of pressure and temperature of the landfill gas.

Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	EE_y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	-
Description of measurement methods and procedures to be applied	As per paragraph 21 of AMS.III.G. (Version 10.0) Specification provided by the equipment manufacturer. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for biogas. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
Frequency of monitoring/recording	-
Value applied	40%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

5.3 Monitoring Plan

1. The requirement of monitoring plan

The project participants will monitor the emission reductions by methods, indicators, and frequency required by Monitoring Methodology AMS-III.G (Version 10.0) and AMS-I.D (Version 18.0) to ensure project ERs are measurable and real. The monitoring methodology is based on direct measurement of the amount of landfill gas captured and destroyed by the project and electricity generating units.

2. Responsibilities of operational and management structure

The project participant will implement this monitoring plan. The plan could be revised according to suggestions from DOE and the practical circumstances, in order to keep it consistent, transparent and conservative during the monitoring process.

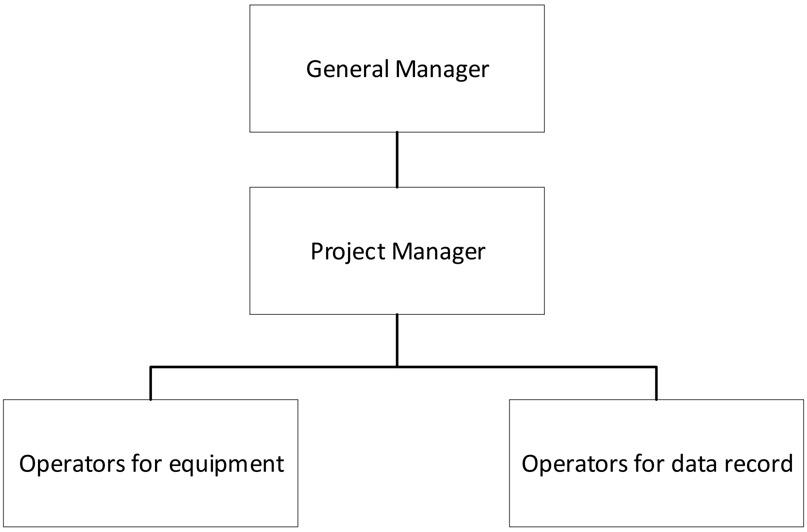


Figure 5.1 Operational and management structure

(1) Principal of the monitoring procedure

The general manager of the project is the leader of the monitoring tasks: he sets out the responsibility of everyone in the monitoring system, and establishes the related documents. Thegeneral manager ensures that staff in the monitoring system has the ability to deal with theassigned tasks.

(1) Executive person of the monitoring procedure

Project Manager: a Project Manager is appointed, specifically responsible for training, checkingthe daily operation, reporting forms and archiving emergency situation reports. The ProjectManager reports monthly to the General Manager (GM) about the project performance andmonitored data. In the event that non-conformance in the performance to the mentioned procedures and/or functioning problems of the monitoring equipment are identified, the ProjectManager will inform the GM about the situation and work out relevant measures to be taken. TheProject Manager will also be responsible for aggregating the monitored data monthly and yearly,archiving and keeping data during the crediting period and two years after.

(2) Operators of the monitoring procedure

Operators will take turns to work in the control center 24 hours a day. They will be in charge of data supervision, filling operation report forms and checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency situation reports.

3. Monitoring system

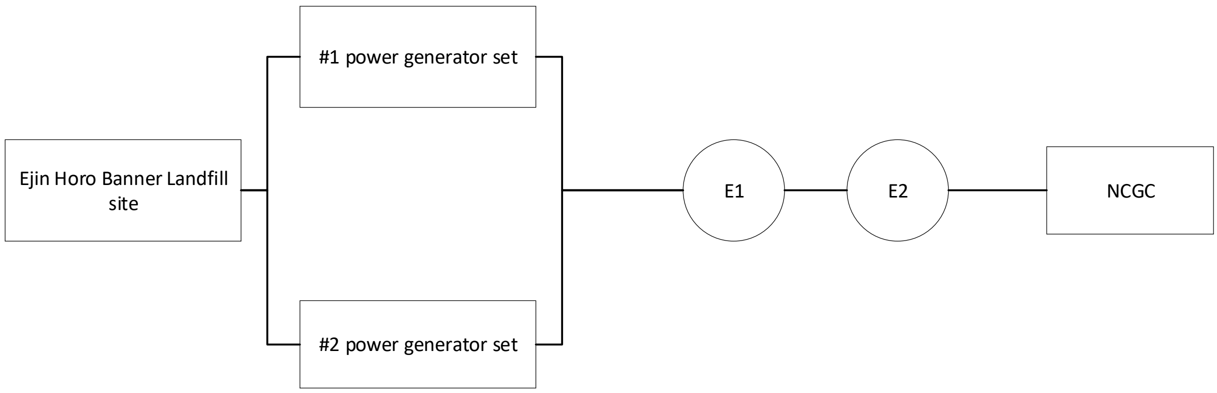


Figure 5.2 Monitoring system

Main parameters:

E1	Electricity meter (accuracy:0.5S) to continuously measure the total electricity by all the generators
E2	Electricity meter (main meter, bi-directional, accuracy:0.5S) to continuously measure the electricity supplied to the NCGC and the electricity supplied by the grid

4.QA/QC

The operator who is responsible for equipment maintenance invites a qualified third party to calibrate the meters (flow meters, gas analyzer and electricity meters). The calibration reports were checked by the project manager and then archived in the project owner archives by the operator. When the data was not available from the main monitoring devices, the data measured by the back-up devices or estimation based on the historical data was used.

The project performance and monitoring data were reported monthly to the General Manager for review and internal audit of calculation and archived and kept for emission reduction calculation yearly during the crediting period and two years after.

5.Emergency procedures

The mistakes occurred in daily operation include problems of the equipment and data. If problems were found and/or monitored, each department reports to the Operation Department, who asked the corrections by providing some suggestions.

As for the data acquisition system, the DCS consists of monitoring and recording the data. Once the system had some problems, the Technical Department and/or factory technicians solved as soon as possible. The DCS data during this period was not adopted.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification
/	/	/